

Experiments continued

What matters for compatibility recovery?

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Recap

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Where we stopped last time

Key points

- Metadata routing changes expert selection but does not reliably improve utility
- Learned proxy models and metadata-conditioned variants did not recover utility-aligned rankings
- The bottleneck was localized to compatibility estimation, not the routing mechanism itself

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The new question

How can we estimate compatibility without leakage?

I tested three directions:

- Learned utility predictors
- Support-free AE utility calibrator
- Target-local support-set NELBO

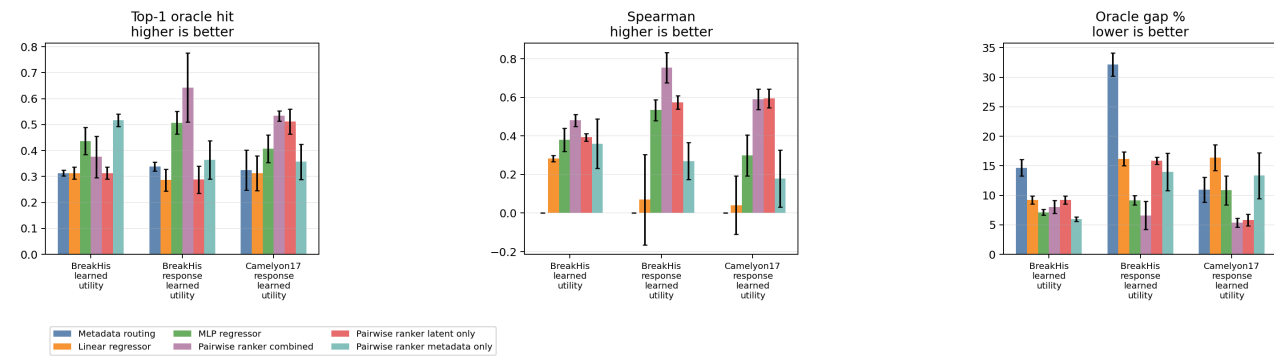
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Results

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Result A

Learned utility is not yet adoption-ready



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Result B

Support-free AE utility calibration

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The idea is: each source domain trains its own Autoencoder on frozen feature embeddings. At test time, a target/query sample is passed through every source AE. If source AE e reconstructs the query embedding well, the query is assumed to lie near source domain e 's learned feature manifold.

Result C

Support-Set NELBO is a strong positive result

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